

Variants of Naive Bayes

- ① Bernoulli Naive Bayes
- ② Multinomial Naive Bayes
- ③ Gaussian Naive Bayes

① Bernoulli Naive Bayes

Whenever your feature are following a Bernoulli Distribution, then we need to use Bernoulli Naive Bayes Algorithm.

Bernoulli $\rightarrow 0, 1$

Data set

f_1	f_2	f_3	o/p
yes	Pass	Male	Yes
yes	Fail	Female	No
No	Pass	Male	yes
yes	Fail	Female	No

① Multinomial Naive Bayes $\Rightarrow$ I/P = Text

Dataset :- Spam Classification

I/P $\rightarrow$ Email Body O/P
 Feature Spam/Not Spam

You have won million \$
 \$ Lottery Spam

DHEERAJ YOU HAVE
 DONE GOOD JOB HAM



Numerical values $\Rightarrow$ Natural Language Processing



Vectors

① Bow

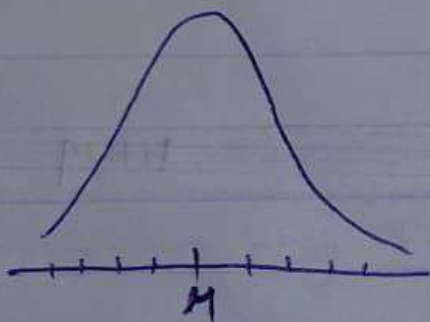
② TF-IDF

③ Word2Vec

③ Gaussian Naive Bayes

If the features are following Gaussian Distribution, then we are Gaussian Naive Bayes

IRIS Dataset



DATASET $\rightarrow$ CONTINUOUS

Age	Height	Weight	yes/no
25	170	78	
38	160	75	
32	150	60	
34	140	35	